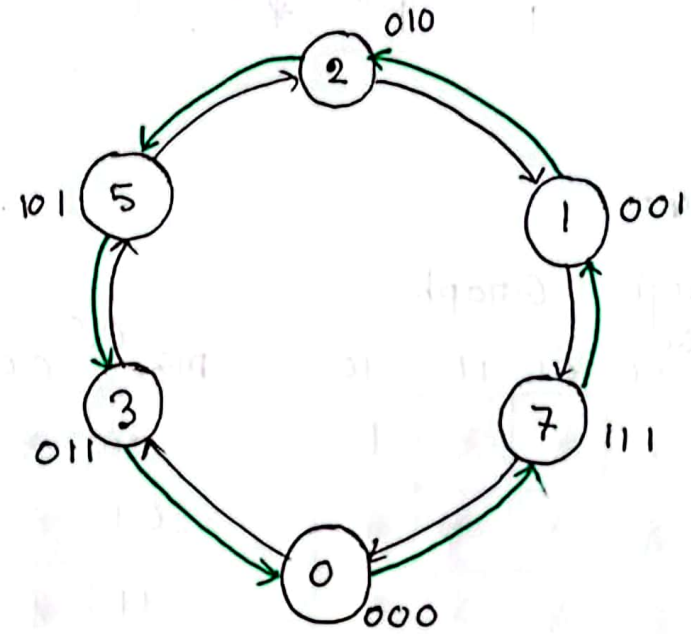
Assignment -02 : Design a irregular synchronous up-down counter for the equation sequence

2 - 1 - 7 - 0 - 3 - 5

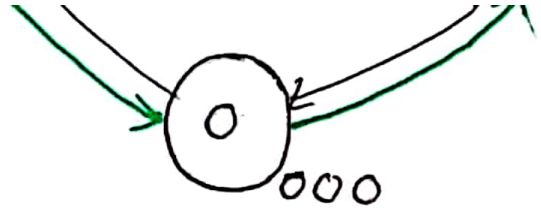
Solution :

Step-01 : State Diagram,



Step-02 : Next State Level :

Present State			Next State, M = 0 (UP)			Next State, M = 1 (Down)		
Q_2^t	Q_1^t	Q_0^t	Q_2^{t+1}	Q_1^{t+1}	Q_0^{t+1}	Q_2^{t+1}	Q_1^{t+1}	Q_0^{t+1}
0	1	0	0	0	1	1	0	1
0	0	1	1	1	1	0	1	0
1	1	1	0	0	0	0	0	1
0	0	0	0	1	1	1	1	1
0	1	1	1	0	1	0	0	0
1	0	1	0	1	0	0	1	1



Step-02 : Next State Level :

Present State			Next State, M = 0 (UP)			Next State, M = 1 (Down)		
Q_2^t	Q_1^t	Q_0^t	Q_2^{t+1}	Q_1^{t+1}	Q_0^{t+1}	Q_2^{t+1}	Q_1^{t+1}	Q_0^{t+1}
0	1	0	0	0	1	1	0	1
0	0	1	1	1	1	0	1	0
1	1	1	0	0	0	0	0	1
0	0	0	0	1	1	1	1	1
0	↑	1	1	0	1	0	0	0
1	0	1	0	1	0	0	1	1

CamScanner

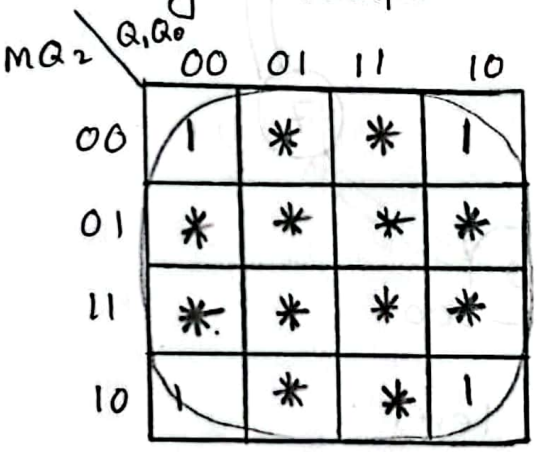
Step-03:

Transition table for the J-K flip-flop

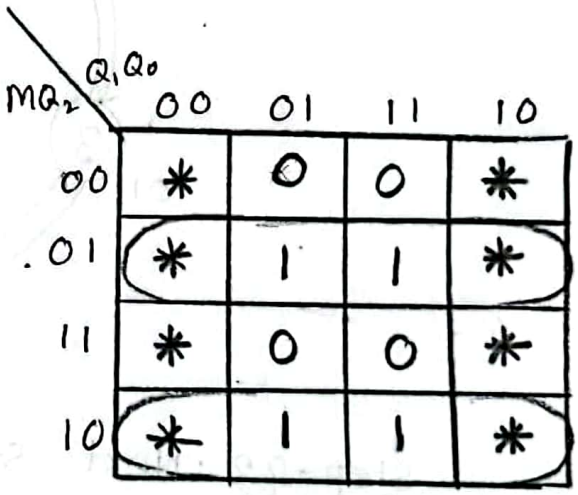
Q^t	Q^{t+1}	J	K
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

Step-04:

Karnaugh Graph

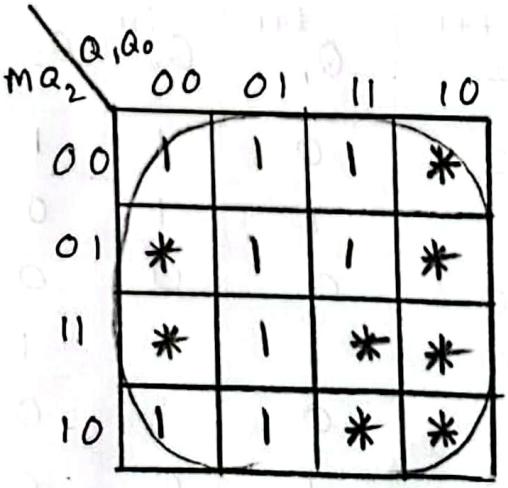


$J_0 = 1$

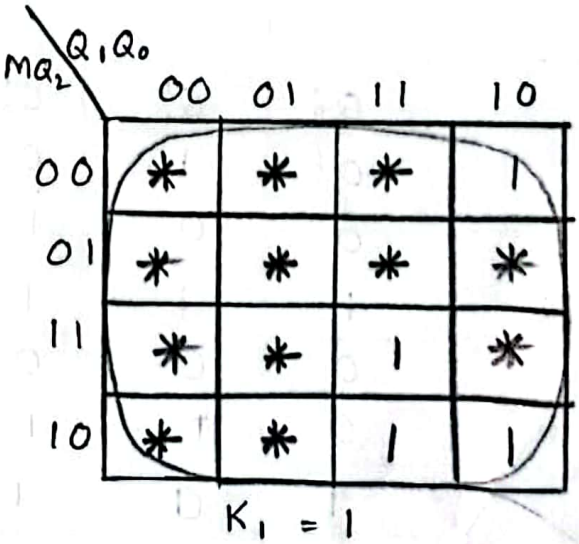


$K_0 = \bar{M}Q_2 + M\bar{Q}_2$

$= M \oplus Q_2$



$J_1 = 1$



$K_1 = 1$

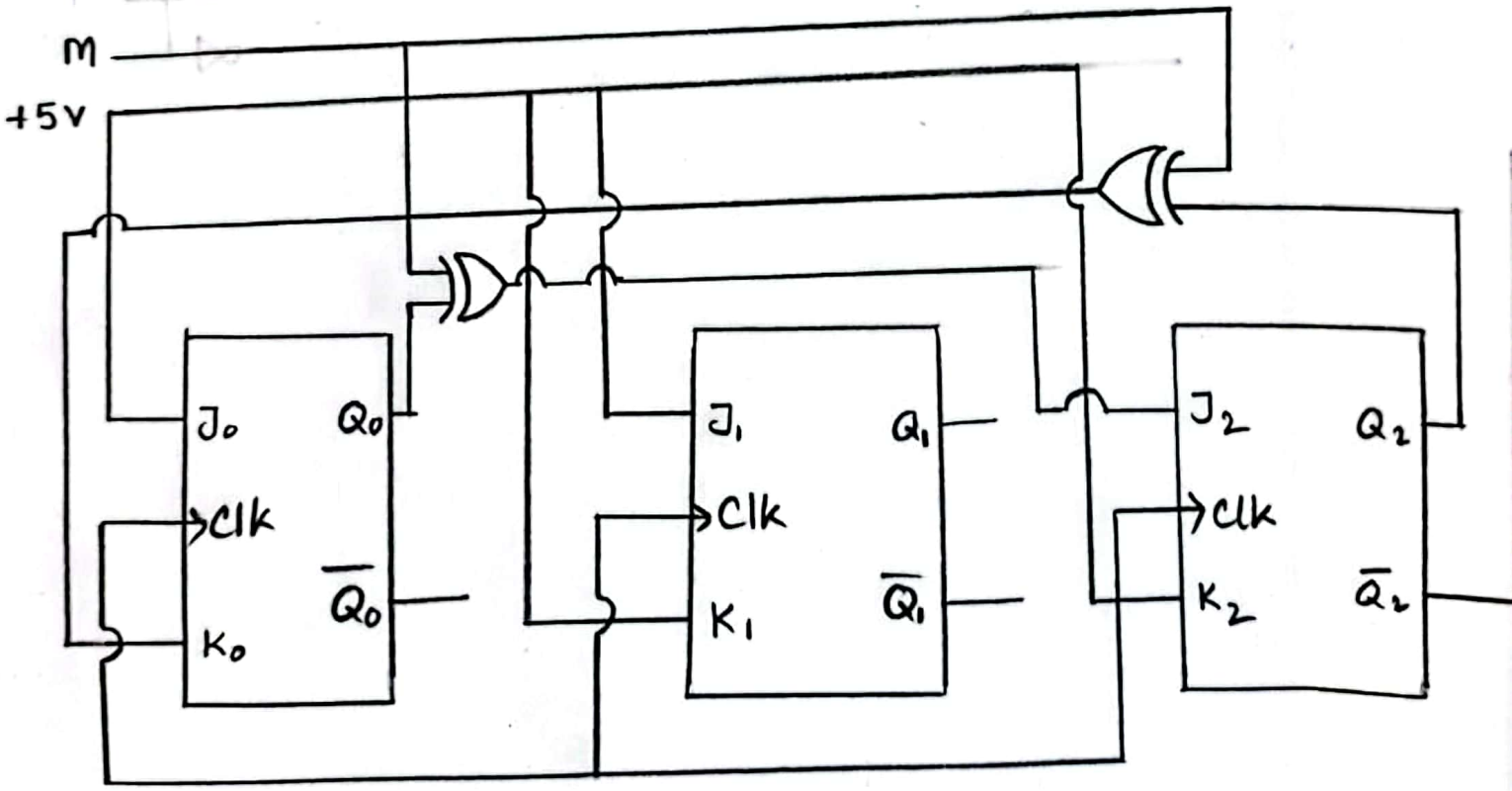
MQ_2 \ Q_1Q_0	00	01	11	10
00	0	1	1	0
01	*	*	*	*
11	*	*	*	*
10	1	0	0	1

$$J_2 = Q_0\overline{M} + \overline{Q_0}M$$

$$= Q_0 \oplus M$$

MQ_2 \ Q_1Q_0	00	01	11	10
00	*	*	*	*
01	*	1	1	*
11	*	1	1	*
10	*	*	*	*

$$K_2 = 1$$



CamScanner